

3.3.1 Derivatives Worksheets

****Remember that simplifying before attempting to find the derivative is worthwhile****

1. Using Leibniz notation $\left(i.e., \frac{dy}{dx}, \frac{dP}{dt}, etc.\right)$, find the derivatives for each of the following functions.

a) $y = 4x^3 + e^3$

b) $y = x^{5/3} - x^{2/3}$

c) $A(s) = -\frac{12}{s^5} + \pi^2$

d) $g(u) = \frac{u^2 + 4u + 3}{\sqrt{u}}$

e) $w(r) = \sqrt[3]{r}(r-1) + 3e^r$

f) $h(t) = (t-2)(2t+3)$

2. Find the equation of the tangent line to the curve at the given point.

a) $y = x^4 + 2x^2 - x$, $(1, 2)$

b) $y = (1 + 2x)^2$, $(1, 9)$

c) $y = x + \sqrt{x}$, $(1, 2)$

3. Consider the function $y = x^2(1 - 2x) + e$.

a) Determine all value of x at which the function has horizontal tangent lines.

b) Determine all value of x at which the function has tangent lines with slope -8 .